

1. Write a script to perform the following three operations on given list Access the third element of a list List Length: Print the total number of items Check if the list is empty Given Input: numbers = [10, 20, 30, 40, 50]

```
numbers = [10, 20, 30, 40, 50]

print(numbers[2])          # third element
print(len(numbers))        # list length
print(len(numbers) == 0)   # check if empty

30
5
False
```

2. Take a given list and modify it through five specific actions: Change Element: Change the second element of a list to 200 and print the updated list. Append Element: Add 600 o the end of a list and print the new list. Insert Element: Insert 300 at the third position (index 2) of a list and print the result. Remove Element (by value): Remove 600 from the list and print the list. Remove Element (by index): Remove the element at index 0 from the list print the list. Given Input: Initial List: [100, 50, 400, 500]

```
my_list = [100, 50, 400, 500]

my_list[1] = 200
print(my_list)

my_list.append(600)
print(my_list)

my_list.insert(2, 300)
print(my_list)

my_list.remove(600)
print(my_list)

del my_list[0]
print(my_list)

[100, 200, 400, 500]
[100, 200, 400, 500, 600]
[100, 200, 300, 400, 500, 600]
[100, 200, 300, 400, 500]
[200, 300, 400, 500]
```

3. Calculate the total sum of all integers in a list and find the arithmetic mean (average). Given Input: Numbers: [10, 20, 30, 40, 50]

```
numbers = [10, 20, 30, 40, 50]

total = sum(numbers)
average = total / len(numbers)

print("Sum:", total)
print("Average:", average)

Sum: 150
Average: 30.0
```

4. Identify the largest and smallest numerical values within a provided list. Given Input: Data: [45, 12, 89, 2, 67]

```
data = [45, 12, 89, 2, 67]

print("Largest:", max(data))
print("Smallest:", min(data))
```

```
Largest: 89
Smallest: 2
```

5. Given a list of integers, iterate through the items and count how many are even and how many are odd. Given Input: Numbers: [10, 21, 4, 45, 66, 93, 11]

```
numbers = [10, 21, 4, 45, 66, 93, 11]

even_count = 0
odd_count = 0

for n in numbers:
    if n % 2 == 0:
        even_count += 1
    else:
        odd_count += 1

print("Even:", even_count)
print("Odd:", odd_count)
```

```
Even: 3
Odd: 4
```

6. Write a Python program to create a set, add a new element to it, remove an element using remove(), and discard an element using discard(). Given Input: fruits = {"apple", "banana", "cherry"}

```
fruits = {"apple", "banana", "cherry"}

fruits.add("mango")
print(fruits)

fruits.remove("banana")
print(fruits)

fruits.discard("kiwi")
print(fruits)
```

```
{'banana', 'mango', 'cherry', 'apple'}
{'mango', 'cherry', 'apple'}
{'mango', 'cherry', 'apple'}
```

7. Write a Python program to remove all elements from a set using the variable itself intact. .clear Given Input: colors = {"red", "green", "blue"}

```
colors = {"red", "green", "blue"}

colors.clear()
print(colors)
```

```
set()
```

8. Write a Python program to determine how many elements are in a set without using the built-in len() function. Given Input: animals = {"cat", "dog", "bird", "fish"}

```
animals = {"cat", "dog", "bird", "fish"}

count = 0
for animal in animals:
    count += 1

print("Number of animals:", count)
```

Number of animals: 4

9. Write a Python program to check whether a set is empty using conditional logic, and print an appropriate message based on the result. Given Input: data = set()

```
data = set()

if not data:
    print("The set is empty")
else:
    print("The set is not empty")
```

The set is empty

10. Write a Python program to combine two sets into one, containing all unique elements from both sets. Given Input: set_a = {1, 2, 3, 4} and set_b = {3, 4, 5, 6}

```
set_a = {1, 2, 3, 4}
set_b = {3, 4, 5, 6}

combined = set_a.union(set_b)
print(combined)
```

{1, 2, 3, 4, 5, 6}

11. Write a Python program to add a new key-value pair to a dictionary, modify an existing value, and access a specific key. Given Input: student = {"name": "Alice", "age": 20, "grade": "B"}

```
student = {"name": "Alice", "age": 20, "grade": "B"}

student["city"] = "New York"
student["age"] = 21

print(student["name"])
print(student)
```

Alice
{'name': 'Alice', 'age': 21, 'grade': 'B', 'city': 'New York'}

12. Write a Python program to remove a specific key from a dictionary, retrieve all key-value pairs, and check whether a given key exists. Given Input: car = {"brand": "Toyota", "model": "Camry", "year": 2022, "color": "blue"}

```
car = {"brand": "Toyota", "model": "Camry", "year": 2022, "color": "blue"}

del car["color"]
print(car.items())
print("brand" in car)

dict_items([('brand', 'Toyota'), ('model', 'Camry'), ('year', 2022)])
True
```

13. Write a Python program to create a dictionary by mapping two equal-length lists, one containing keys and the other containing values. Given Input: keys = ["name", "age", "city"] and values = ["Bob", 25, "London"]

```
keys = ["name", "age", "city"]
values = ["Bob", 25, "London"]

result = dict(zip(keys, values))
print(result)

{'name': 'Bob', 'age': 25, 'city': 'London'}
```

14. Write a Python program to remove all items from a dictionary while keeping the dictionary object itself intact. Given Input: inventory = {"apples": 10, "bananas": 5, "oranges": 8}

```
inventory = {"apples": 10, "bananas": 5, "oranges": 8}

inventory.clear()
print(inventory)

{}
```

15. Write a Python program to combine two dictionaries into a single dictionary. If both dictionaries share a key, the value from the second dictionary should take precedence. Given Input: dict1 = {"a": 1, "b": 2} and dict2 = {"b": 3, "c": 4}

```
dict1 = {"a": 1, "b": 2}
dict2 = {"b": 3, "c": 4}

combined = {**dict1, **dict2}
print(combined)

{'a': 1, 'b': 3, 'c': 4}
```

16. Write a Python program to create a tuple, access its elements by index, and find its length. Given Input: fruits = ("apple", "banana", "cherry", "date")

```
fruits = ("apple", "banana", "cherry", "date")

print(fruits[0])
print(fruits[2])
print(len(fruits))

apple
cherry
4
```

17. Write a Python program to create a tuple containing a single item, the number 50, and confirm its type. Given Input: A single integer value 50

```
single = (50,)

print(single)
print(type(single))

(50,)
<class 'tuple'>
```

18. Write a Python program to join three separate tuples into one new tuple using the + operator. Given Input: a = (1, 2), b = (3, 4), and c = (5, 6)

```
a = (1, 2)
b = (3, 4)
c = (5, 6)

joined = a + b + c
print(joined)

(1, 2, 3, 4, 5, 6)
```

19. Write a Python program to reverse the order of elements in a tuple. Given Input: items = (1, 2, 3, 4, 5)

```
items = (1, 2, 3, 4, 5)

reversed_items = items[::-1]
print(reversed_items)

(5, 4, 3, 2, 1)
```

20. Write a Python program to convert a tuple of characters into a single joined string. Given Input: chars = ('a', 'b', 'c')

```
chars = ('a', 'b', 'c')

word = "".join(chars)
print(word)

abc
```